

Fatty Acids

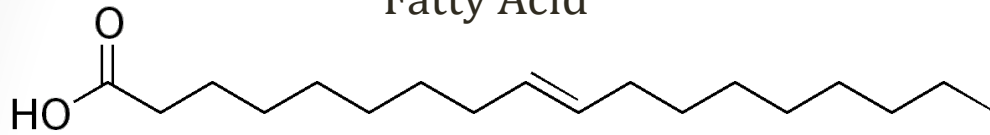
Jason Ryan, MD, MPH

Lipids

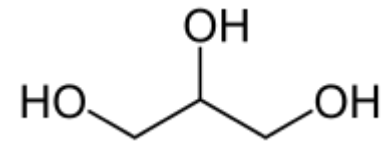
- Mostly carbon and hydrogen
- Not soluble in water
- Many types:
 - Fatty acids
 - Triacylglycerol (triglycerides)
 - Cholesterol
 - Phospholipids
 - Steroids
 - Glycolipids

Lipids

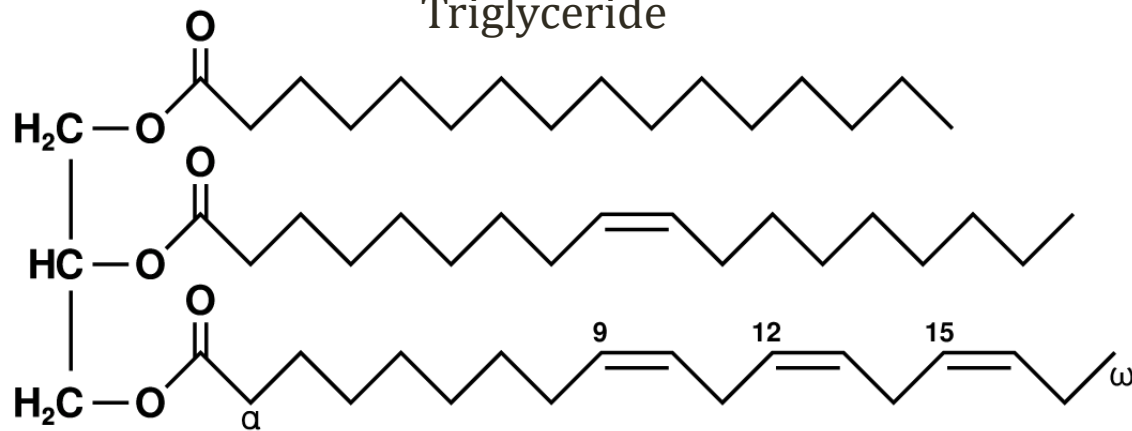
Fatty Acid



Glycerol



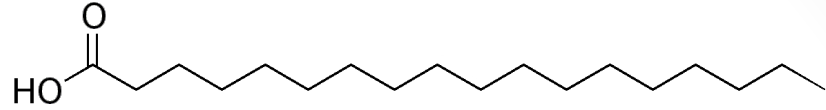
Triglyceride



Fatty Acid and Triglycerides

- Most lipids degraded to **free fatty acids** in intestine
- Enterocytes convert FAs to **triacylglycerol**
- **Chylomicrons** carry through plasma
- TAG degraded back to free fatty acids
 - **Lipoprotein lipase**
 - Endothelial surfaces of capillaries
 - Abundant in adipocytes and muscle tissue

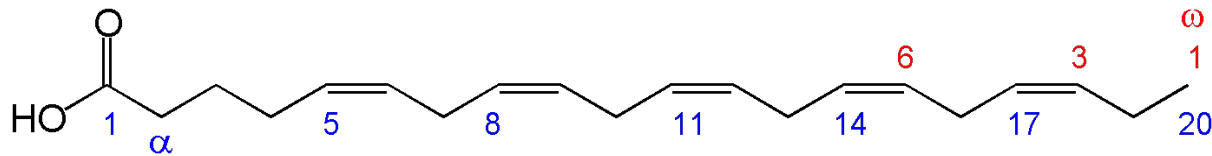
Vocabulary



- “Saturated” fat (or fatty acid)
 - Contains **no double bonds**
 - “Saturated” with hydrogen
 - Usually solid at room temperature
 - Raise LDL cholesterol
- “Unsaturated” fat
 - Contains **at least one double bond**
- “Monounsaturated:” One double bond
- “Polyunsaturated:” More than one double bond

More Vocabulary

- Trans fat
 - Double bonds (unsaturated) can be trans or cis
 - Most natural fats have cis configuration
 - Trans from partial hydrogenation (food processing method)
 - Can increase LDL, lower HDL
- Omega-3 fatty acids
 - Type of polyunsaturated fat
 - Found in fish oil
 - Lower triglyceride levels



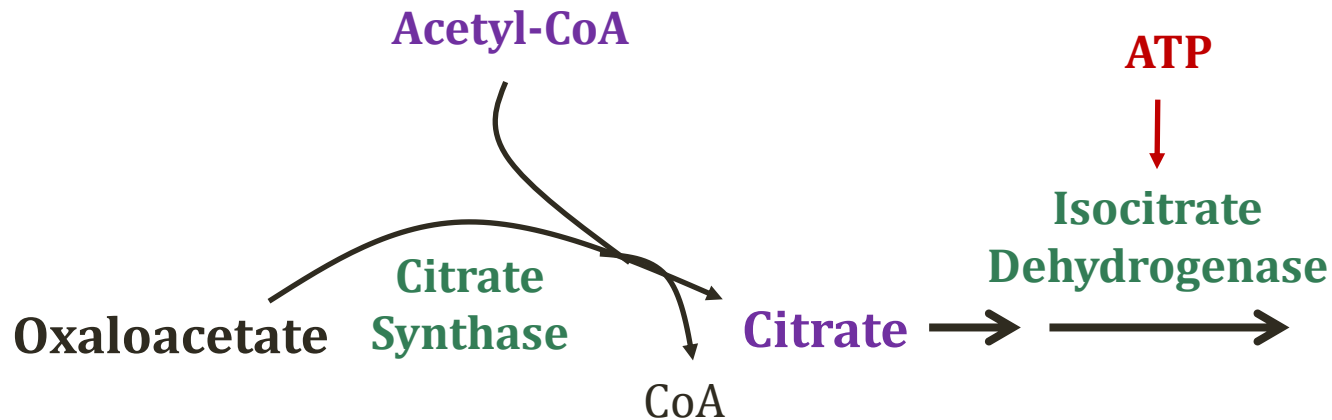
eicosapentaenoic acid (EPA)

Fatty Acid Metabolism

- Fatty acids **synthesis**
 - Liver, mammary glands, adipose tissue (small amount)
 - Excess carbohydrates and proteins → fatty acids
- Fatty acid **storage**
 - Adipose tissue
 - Stored as triglycerides
- Fatty acid breakdown
 - **β-oxidation**
 - Acetyl CoA → TCA cycle → ATP

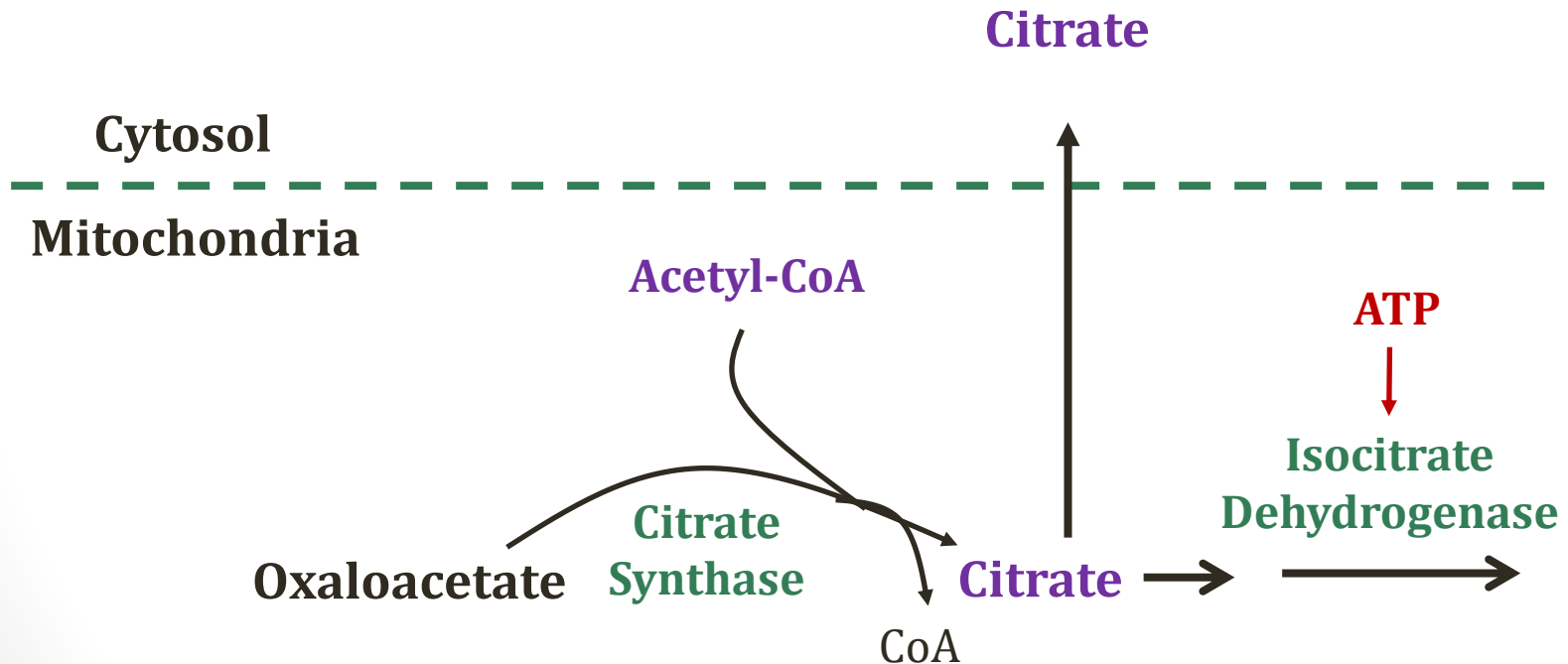
Fatty Acid Synthesis

- In high energy states (fed state):
 - Lots of acetyl-CoA
 - Lots of ATP
 - Inhibition of isocitrate dehydrogenase (TCA cycle)
- Result: **High citrate level**



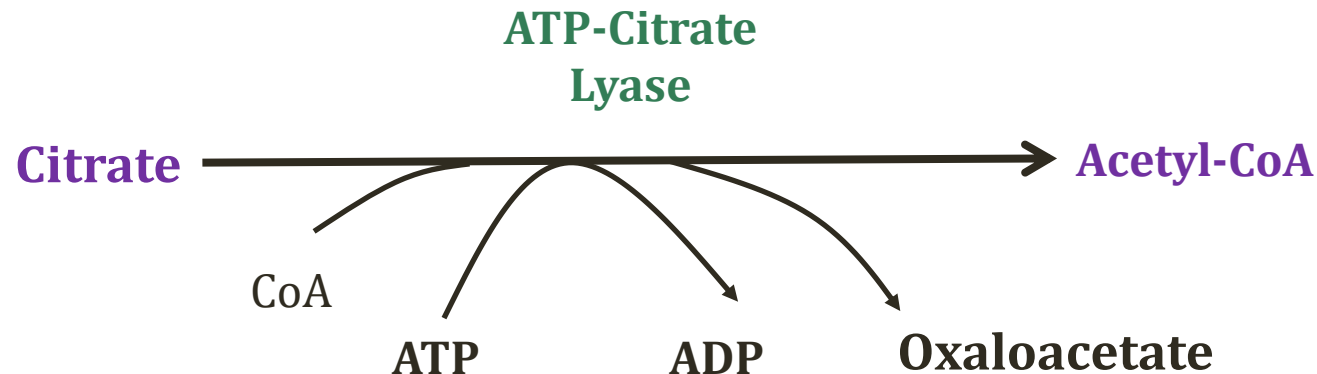
Fatty Acid Synthesis

- Step 1: Citrate to cytosol via citrate shuttle
- Key point: Acetyl-CoA cannot cross membrane



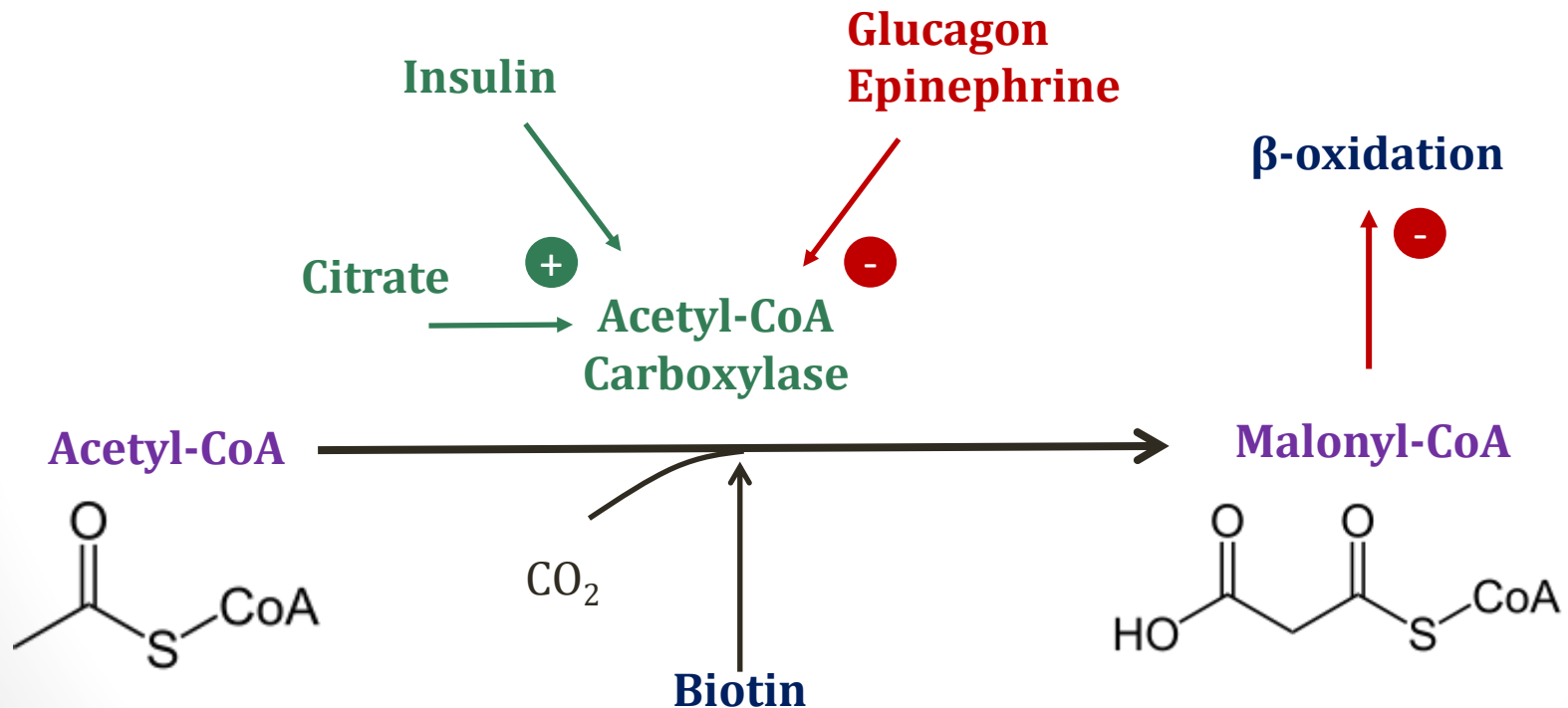
Fatty Acid Synthesis

- Step 2: Citrate converted to acetyl-CoA
- Net effect: Excess acetyl-CoA moved to cytosol



Fatty Acid Synthesis

- Step 3: Acetyl-CoA converted to malonyl-CoA
- Rate limiting step

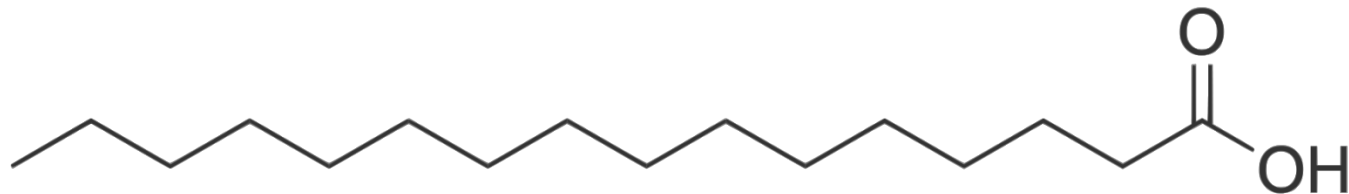


Biotin

- Cofactor for carboxylation enzymes
 - All **add 1-carbon group** via CO_2
 - Pyruvate carboxylase
 - **Acetyl-CoA carboxylase**
 - Propionyl-CoA carboxylase

Fatty Acid Synthesis

- Step #4: Synthesis of palmitate
- Enzyme: **fatty acid synthase**
- Uses carbons from acetyl CoA and malonyl CoA
- Creates **16 carbon** fatty acid
- Requires **NADPH** (HMP Shunt)



Palmitate

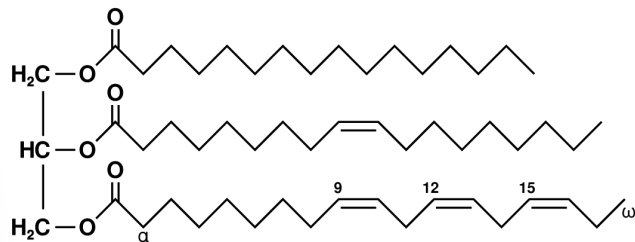
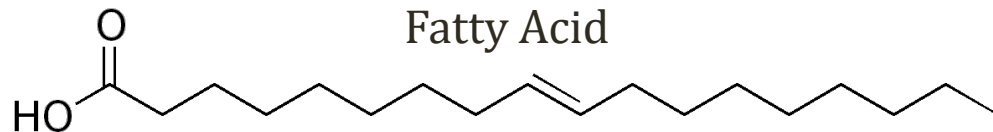
Fatty Acid Storage

- Palmitate can be modified to other fatty acids
- Used by various tissues based on needs
- Stored as triacylglycerols in adipose tissue

Fatty Acid Breakdown

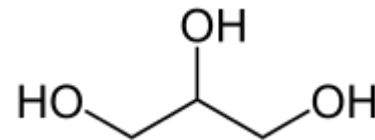
- Key enzyme: **Hormone sensitive lipase**
- Removes fatty acids from TAG in adipocytes
- Activated by **glucagon** and **epinephrine**

Fatty Acid Breakdown



Triacylglycerol

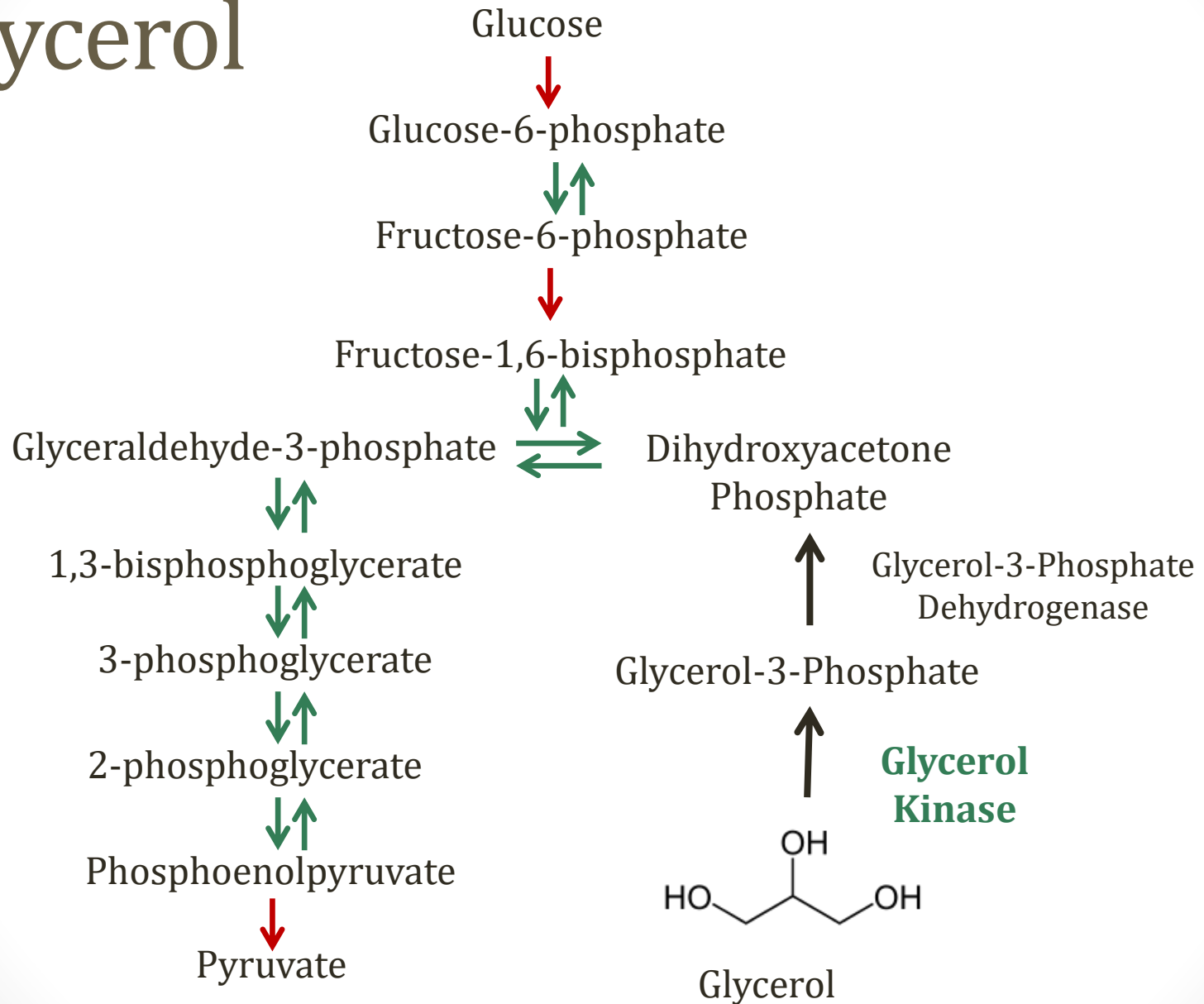
Hormone
Sensitive
Lipase



Glycerol

→ Liver

Glycerol

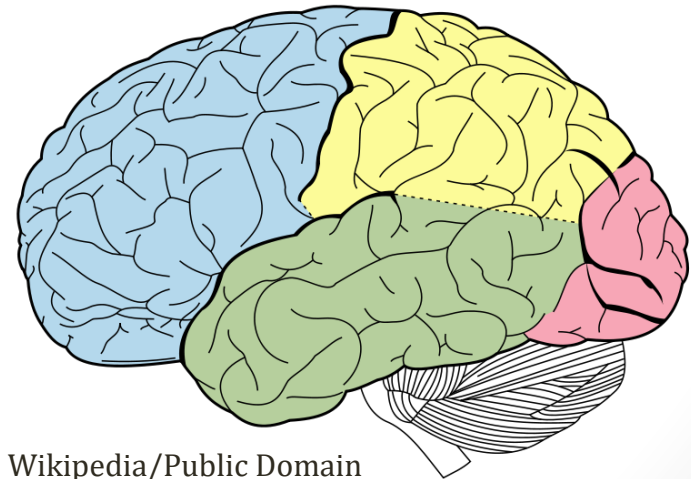


Fatty Acid Breakdown

- Fatty acids transported via **albumin**
- Taken up by tissues
- Not used by:
 - **RBCs**: Glycolysis only (no mitochondria)
 - **Brain**: Glucose and ketones only



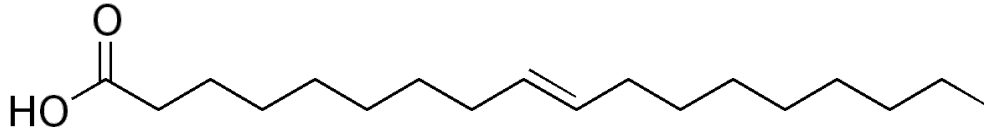
Database Center for Life Science (DBCLS)



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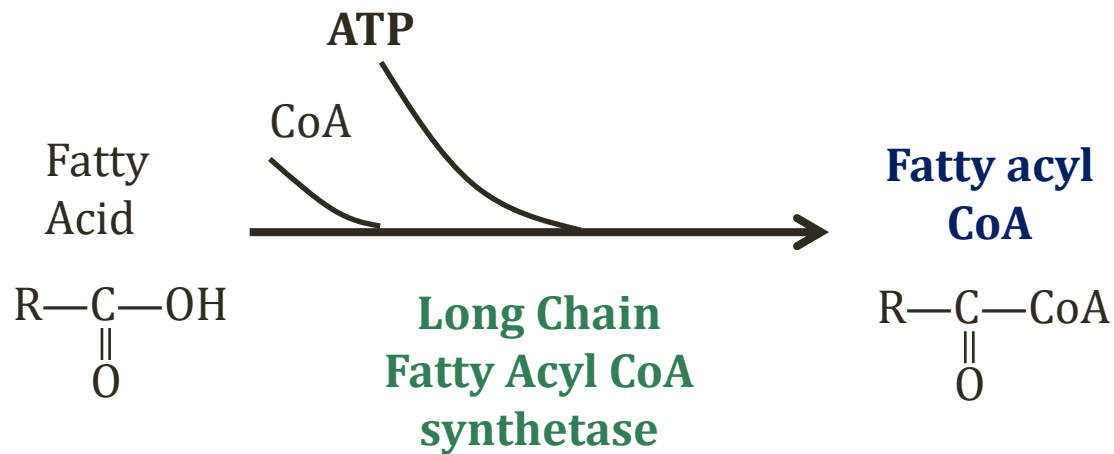
Fatty Acid Breakdown

- **β -oxidation**
- Removal of 2-carbon units from fatty acids
- Produces acetyl-CoA, NADH, FADH₂



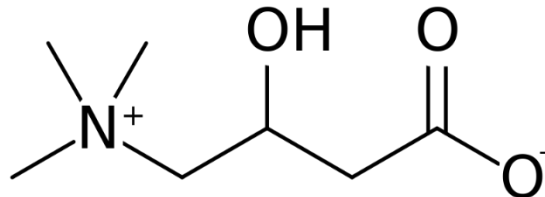
β -oxidation

- Step #1: Convert fatty acid to fatty acyl CoA



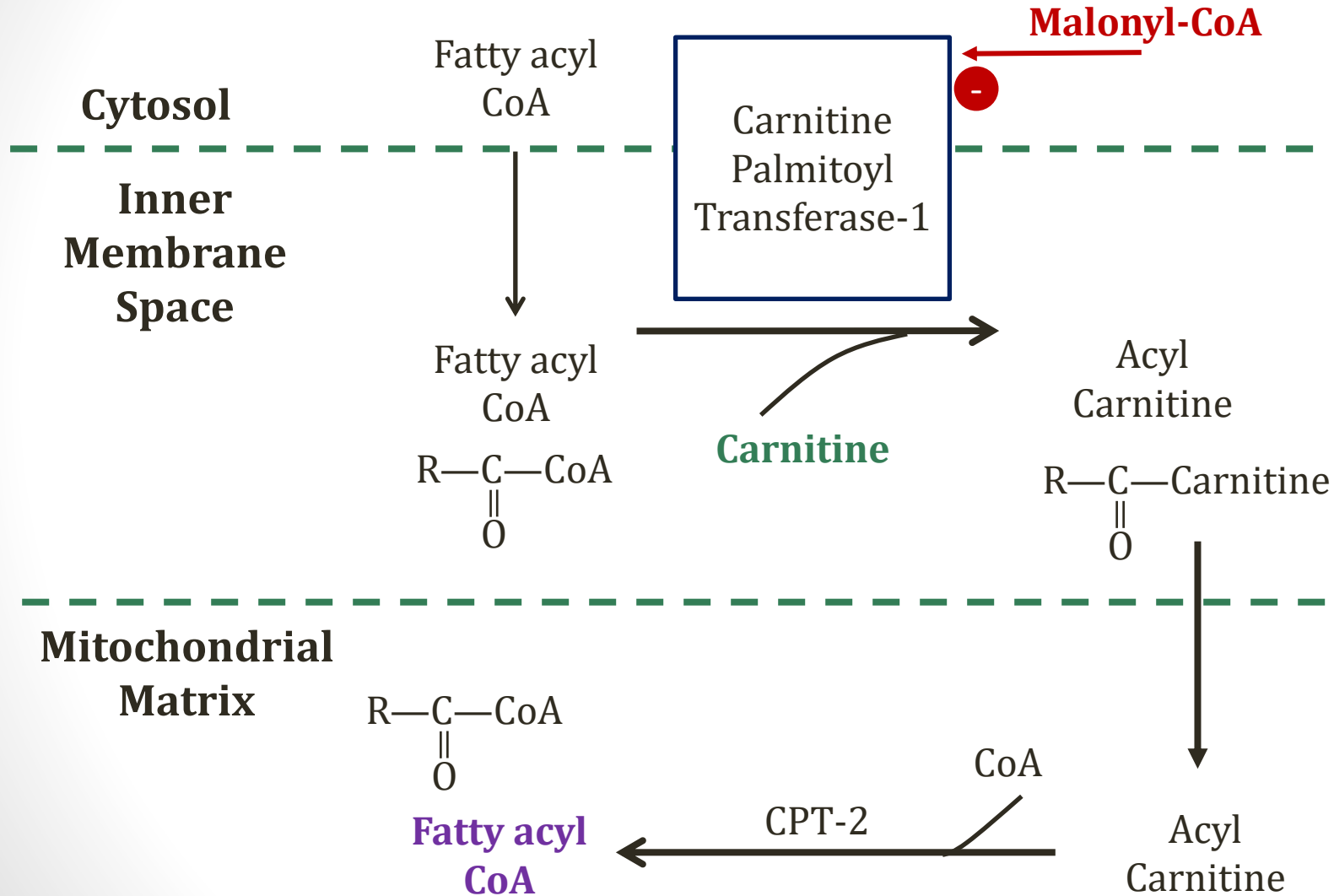
β -oxidation

- Step #2: Transport fatty acyl CoA $\rightarrow$ inner mitochondria
- Uses **carnitine shuttle**
- Carnitine in diet
- Also synthesized from lysine and methionine
 - **Only liver, kidney** can synthesize de novo
 - **Muscle and heart** depend on diet or other tissues



Carnitine

Carnitine Shuttle



Carnitine Deficiencies

- Several potential secondary causes
 - Malnutrition
 - Liver disease
 - Increased requirements (trauma, burns, pregnancy)
 - Hemodialysis (↓ synthesis; loss through membranes)
- Major consequence:
 - Inability to transport LCFA to mitochondria
 - Accumulation of LCFA in cells
- Low serum **carnitine** and **acylcarnitine** levels

Carnitine Deficiencies

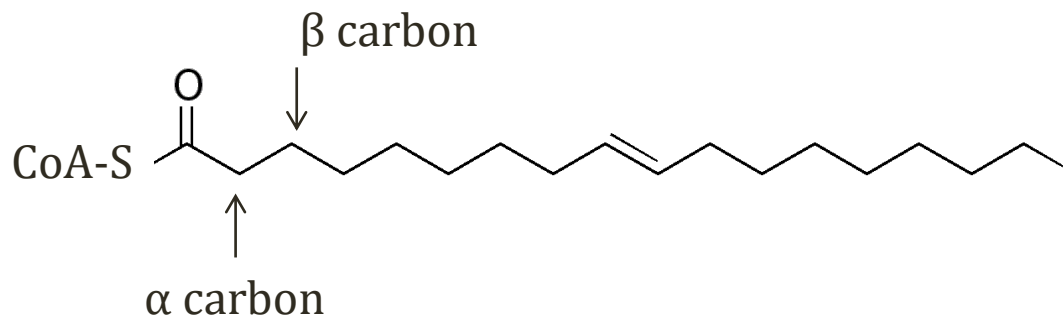
- **Muscle** weakness, especially during exercise
- **Cardiomyopathy**
- **Hypoketotic hypoglycemia** when fasting
 - Tissues overuse glucose
 - Poor ketone synthesis without fatty acid breakdown

Primary systemic carnitine deficiency

- Mutation affecting carnitine **uptake** into cells
- Infantile phenotype presents first two year of life
 - Encephalopathy
 - Hepatomegaly
 - Hyperammonemia (liver dysfunction)
 - Hypoketotic hypoglycemia
 - Low serum carnitine: kidneys cannot resorb carnitine
 - Reduced carnitine levels in muscle, liver, and heart

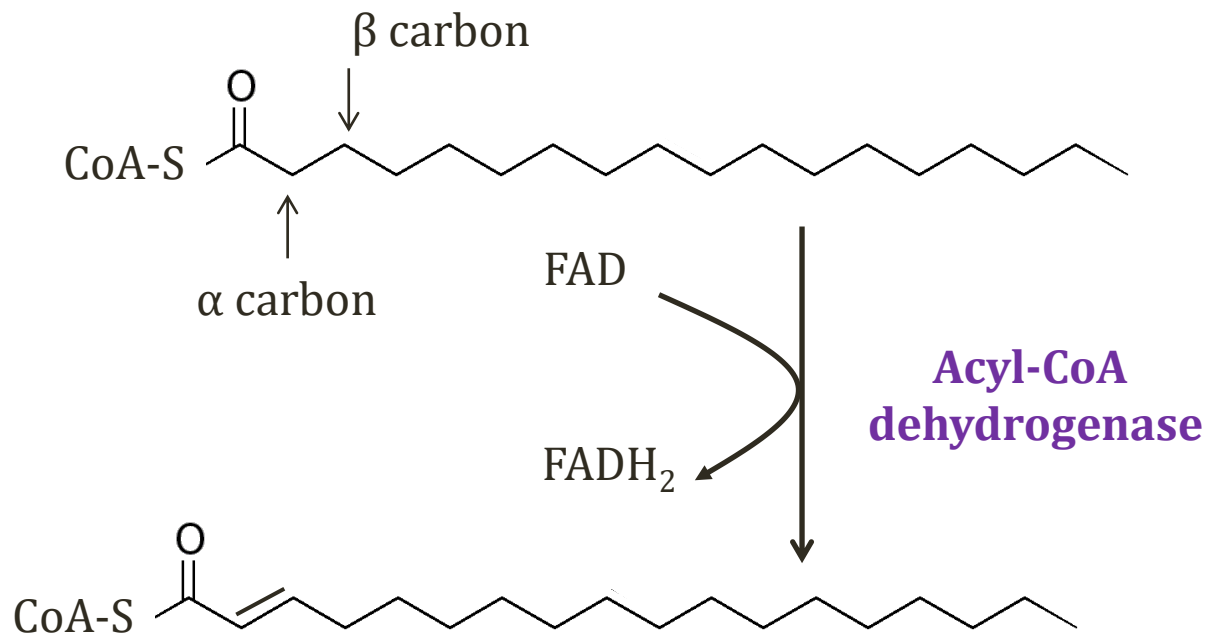
β -oxidation

- Step #3: Begin “cycles” of beta oxidation
- Removes two carbons
- Shortens chain by two
- Generates NADH, FADH₂, Acetyl CoA



β -oxidation

- First step in a cycle involves **acyl-CoA dehydrogenase**
- Adds a double bond between α and β carbons



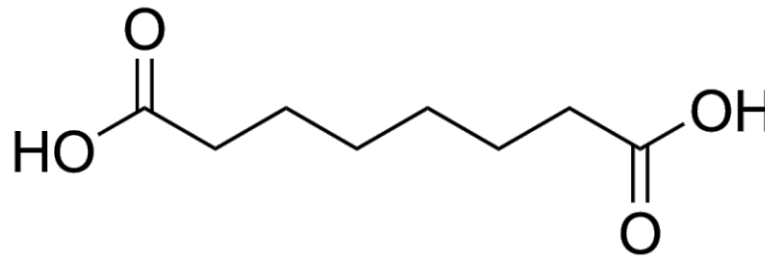
Acyl-CoA Dehydrogenase

- Family of four enzymes
 - Short
 - Medium
 - Long
 - Very-long chain fatty acids
- Well described deficiency of medium chain enzyme

MCAD Deficiency

Medium Chain Acyl-CoA Dehydrogenase

- Autosomal recessive disorder
- Poor oxidation 6-10 carbon fatty acids
- Severe **hypoglycemia without ketones**
- Dicarboxylic acids 6-10 carbons in urine
- **High acylcarnitine levels**



Dicarboxylic Acid

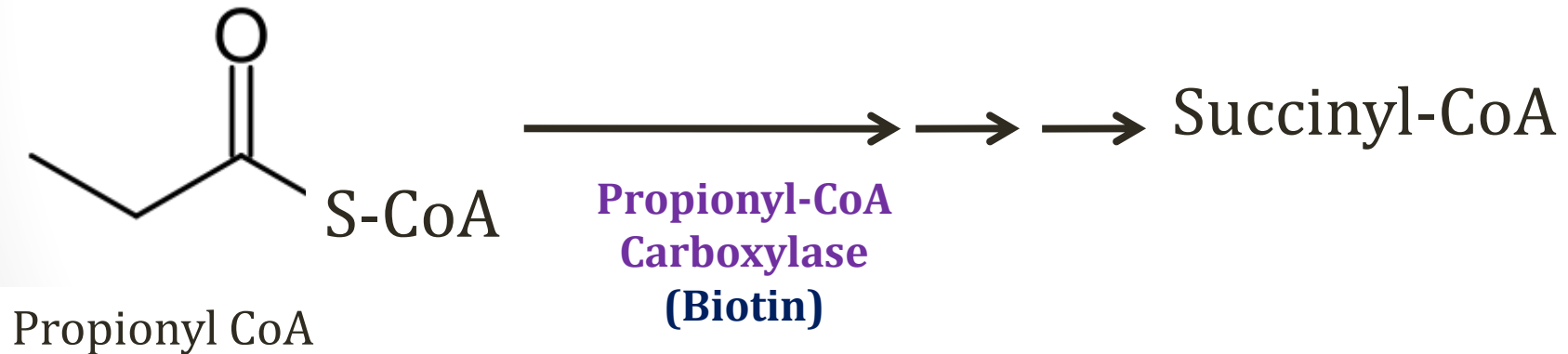
MCAD Deficiency

Medium Chain Acyl-CoA Dehydrogenase

- **Gluconeogenesis shutdown**
 - Pyruvate carboxylase depends on Acetyl-CoA
 - Acetyl-CoA levels low in absence β -oxidation
- Exacerbated in **fasting/infection**
- Treatment: Avoid fasting

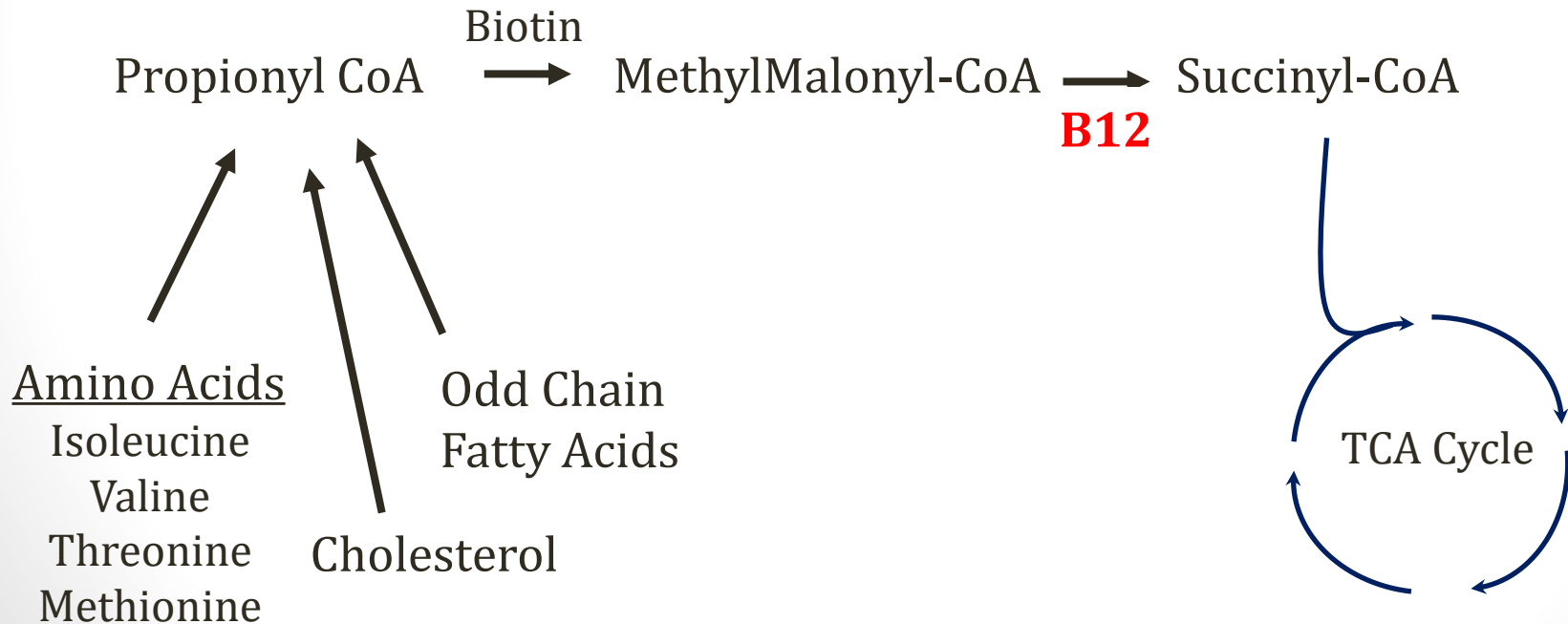
Odd Chain Fatty Acids

- β -oxidation proceeds until 3 carbons remain
- Propionyl-CoA $\rightarrow$ Succinyl-CoA $\rightarrow$ TCA cycle
- Key point: Odd chain FA $\rightarrow$ **glucose**



Propionyl CoA

- Common pathway to TCA cycle
- Elevated methylmalonic acid seen in B12 deficiency



Methylmalonic Acidemia

- Deficiency of **Methylmalonyl-CoA mutase**
- Anion gap metabolic acidosis
- CNS dysfunction
- Often fatal early in life

